

Phu Vinh Nguyen

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EDUCATION

Northeastern University Boston, MA
Certificate in Bioinformatics, concentration in programming May 2025
Relevant Coursework: Python for Bioinformatics, Genomics in Bioinformatics, Algorithms, R programming

Uniwersytet Jagielloński Krakow, PL
Master's in Molecular Biotechnology Jul 2021

University of California, Irvine Irvine, CA
Bachelor of Science in Pharmaceutical Sciences Sept 2019

WORK EXPERIENCE

Lab Automation Engineer | Developer, Sanofi Pasteur Jun 2025 – Present

- Built a cross-platform RESTful application using Electron, React, and Node.js to automate internal request forms and ticketing systems via SMTP integration, serving 100+ R&D users and reducing on-call workloads by 15%.
- Analyzed and managed testing methods for liquid handler robots to improve automation by 10 times conventional means.
- Oversaw biweekly presentations to primary stakeholders communicating performance, current objectives and future improvements.

Automation Engineer, Foundation Medicine Inc Nov 2024 – Apr 2025

- Developed a Python GUI application with CustomTkinter that served as a client interface for backend automation systems, enabling 50+ R&D users to execute complex automation workflows independently and reducing support requests by 20%.
- Built a React application that reads and displays Hamilton robot configuration metadata, providing teams with clear visibility into system parameters and object settings reducing troubleshooting time by 40%.
- Created barcode scanning systems and ETL workflows to integrate sample data into Sapio LIMS, automating sample tracking and reducing manual entry errors.

Lab Automation Engineer, Sanofi Pasteur Oct 2022 – Nov 2024

- Developed a suite of Python scripts to automate routine tasks, including data parsing and data analysis.
- Optimized high-throughput Hamilton liquid handling methods, reducing assay processing times by up to 30% in DNA extraction and fluorescence-based workflows.
- Integrated laboratory automation devices into robotic workflows through TCP/IP and APIs, to enable command execution and data exchange across multiple devices.

Senior Research Associate, Dermibiont Inc. Oct 2021 – Sept 2022

- Established scalable genomic sequencing workflows using Nanopore to build automated data analysis pipelines with open-source bioinformatics tools to support high-throughput microbiome research.
- Collaborated with external bioinformatics teams to troubleshoot and optimize Dockerized QC pipelines, ensuring consistent data quality standards and reducing processing errors by 25% across research projects.

PROJECTS

[Personal Website](#)

- Developed a React portfolio website to showcase professional experience, technical projects, and programming skills.

SKILLS

Languages: Python, JavaScript, SQL, HTML, CSS, Bash, R

Frameworks and Libraries: React, Node.js, Express, Electron, CustomTkinter

Technical Tools: Git, Docker, Jira, LIMS (user level), HPC environments, REST API, SMTP

Database: MySQL, PostgreSQL